THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS UNIT TEST SET C

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The ones digit of a perfect square can never be: 1

(a) 9 **(b)** 4 **(c)** 7 **(d)** 1

2. The sum of the first 20 odd natural numbers is: 1

(a) 399 **(b)** 400 **(c)** 420 **(d)** 40

3. $2^5 \times 2^3$ equals: 1

(a) 2^8 **(b)** 4^8 **(c)** 2^{15} **(d)** 2^2

4. The number 72000 in standard (scientific) form is: 1

(a) 72×10^4 **(b)** 7.2×10^4 **(c)** 0.72×10^4 **(d)** 7.2×10^3

5. Our present system of writing numbers with the digits 0 to 9 originated in: 1

(a) Greece **(b)** India **(c)** Rome **(d)** Egypt

6. The Roman numeral MMXXIV stands for: 1

(a) 2014 **(b)** 2025 **(c)** 2024 **(d)** 2034

7. The decimal number 17 written in base 3 is: 1

(a) 200 **(b)** 121 **(c)** 201 **(d)** 122

8. Three angles of a quadrilateral are 50°, 130° and 90°. The fourth angle is: 1

(a) 90° **(b)** 110° **(c)** 80° **(d)** 100°

9. Diagonals that bisect each other at right angles but are unequal belong to a: 1

(a) rectangle **(b)** rhombus **(c)** trapezium **(d)** kite

10. The diagonals of a rhombus are 16 cm and 30 cm. Its side is: 1

(a) 18 cm **(b)** 17 cm **(c)** 16 cm **(d)** 23 cm

11. Is 7314 divisible by 3? 1

(a) Only if the number is prime **(b)** No

(c) Yes **(d)** Only if the last digit is even

12. The LCM of 12 and 18 is: 1

(a) 48 **(b)** 6 **(c)** 36 **(d)** 216

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. What is a perfect square? Give two examples. 2

14. Write 5^{-3} as a fraction. 2

15. Why is the Roman system not suitable for large calculations? 2

16. List two properties of the diagonals of a rectangle. 2

17. What are co-prime numbers? Give an example. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. Express 64 as the sum of consecutive odd numbers starting with 1. 3

19. Simplify: $\dfrac{5^6 \times 5^2}{5^3}$. 3

20. Write 2499 in Roman numerals and explain the grouping. 3

21. Find the smallest number divisible by 15, 20 and 25. 3

22. Find $\sqrt{4624}$ using prime factorisation. 3

OR

Find the cube root of 17576 by prime factorisation. 3

23. Show that $(a \times b)^{-1} = a^{-1} \times b^{-1}$ for $a = 2,\ b = 5$. 3

OR

Find the value of $\left(2^3\right)^2 \times 2^{-4} \div 2^0$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) State the laws of exponents for $a^m \times a^n$, $a^m \div a^n$ and $(a^m)^n$. 5

(ii) Simplify $\dfrac{2^7 \times 5^4}{10^3}$.

OR

Case study: A laboratory sample has 2×10^9 bacteria. The population triples every hour. 5

(i) How many bacteria are there after 3 hours?

(ii) Express the answer in standard form.

(iii) After how many hours will it first exceed 10^{11}?

25. **Case study:** A rectangular tabletop has sides 150 cm and 80 cm. 5

(i) Find its diagonal.

(ii) Find the perimeter.

(iii) A ribbon is to cover both diagonals; find the length.

OR

(i) Compare a rhombus and a rectangle on sides, angles and diagonals. 5

(ii) State the quadrilateral that is both.

END OF QUESTION PAPER — Total: 50 Marks

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